



BAULKHAM HILLS HIGH SCHOOL

2016

YEAR 11

HALF YEARLY EXAMINATION

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 90 minutes
- Write using black or blue pen
- Black pen is preferred
- Board-approved calculators may be used
- All relevant mathematical reasoning and/or calculations must be shown

Total marks – 58

Section I (Pages 2-3)

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Section II (Pages 4-7)

48 marks

Attempt Questions 11-14

Allow about 1 hour 15 minutes for this section

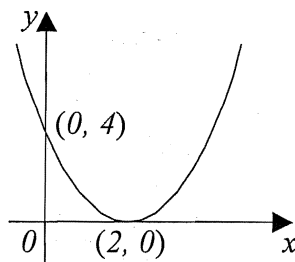
Section I Multiple choice.

10 marks

Allow about 15 minutes for this section.

Use the multiple choice answer sheet for Questions 1-10

- 1) The solution of $\frac{1-3x}{2} \geq 1\frac{1}{2}$ is
(A) $x \leq -\frac{2}{3}$ (B) $x \geq -\frac{5}{3}$ (C) $x \geq -\frac{2}{3}$ (D) $x \leq -\frac{4}{3}$
- 2) What is the domain of the function $y = \frac{\sqrt{5-x}}{x-4}$?
(A) $x < 4$ or $x > 4$
(B) $x \leq 5$
(C) $x < 4$ or $4 < x \leq 5$
(D) $x < 4$ or $x < 5$
- 3) If $\tan A = \frac{2rs}{r^2-s^2}$, where A is acute and $r > s > 0$, then $\cos A$ is equal to
(A) $\frac{\sqrt{r^2-s^2}}{2r}$ (B) $\frac{r^2-s^2}{r^2+s^2}$ (C) $\frac{r}{r-s}$ (D) $\frac{rs}{r^2+s^2}$
- 4) If $(x-3)(2x+1) = 0$ then the possible values of $2x+1$ are
(A) 0 only (B) 0 and 7 (C) $-\frac{1}{2}$ and 3 (D) 0 and 3
- 5)



The diagram could be the graph of

- (A) $y = (x-2)^2 + 4$
(B) $y = x^2 - 2x$
(C) $y = (x-2)^2$
(D) $y = (x+2)^2$

6) The expression $\frac{k}{3}(k+1)(k+2) + (k+1)(k+2)$ is equal to

(A) $\frac{(k+1)(k+3)(k+4)}{6}$

(B) $\frac{(k+1)(k+2)(k+3)}{3}$

(C) $\frac{k(k+1)(k+2)}{3}$

(D) $\frac{2k}{3}(k+1)(k+2)$

7) If $J = K(G - \frac{1}{T})$, then T equals

(A) $\frac{K}{KG - J}$

(B) $\frac{K}{G - J}$

(C) $\frac{K}{J - KG}$

(D) $\frac{J}{K} - G$

8) If $a^2 = a + 2$, then a^3 equals

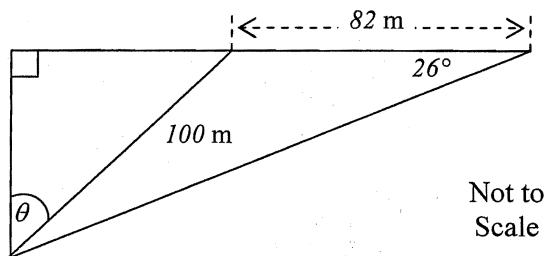
(A) $a + 4$

(B) $2a + 8$

(C) $3a + 2$

(D) $27a + 8$

9) What is the value of θ to the nearest degree?



(A) 21°

(B) 32°

(C) 43°

(D) 55°

10) Solve for x , $\sqrt{7x} - \sqrt{3x} = 4$

(A) $10 - 2\sqrt{21}$

(B) $10 + 2\sqrt{21}$

(C) $\sqrt{7} - \sqrt{3}$

(D) $\sqrt{7} + \sqrt{3}$

END OF SECTION I

SECTION II

48 marks

Attempt questions 11 to 14

Allow about 75 minutes for this section.

Answer each question on the appropriate page of your answer booklet. Each answer sheet must show your name. Extra paper is available.

All necessary working should be shown in every question.

	Marks
Question 11 (12 marks) Answer on the appropriate page of your answer booklet	
(a) Simplify $4x - (2 - x)(1 + x)$	2
(b) Express $\frac{2}{\sqrt{5} - 1}$ with a rational denominator.	2
(c) Solve the equation $4 - \frac{x}{3} = \frac{x}{2}$	2
(d) Factorise $3x^2 - 13x + 12$	2
(e) Solve $2\sin^2\theta = \sin\theta$ for $0 \leq \theta \leq 360^\circ$	2
(f) Consider the function $f(x) = \begin{cases} x^2 & \text{for } x \geq 0 \\ -x^2 & \text{for } x < 0 \end{cases}$	
Determine, without sketching, whether the function is even, odd or neither.	2

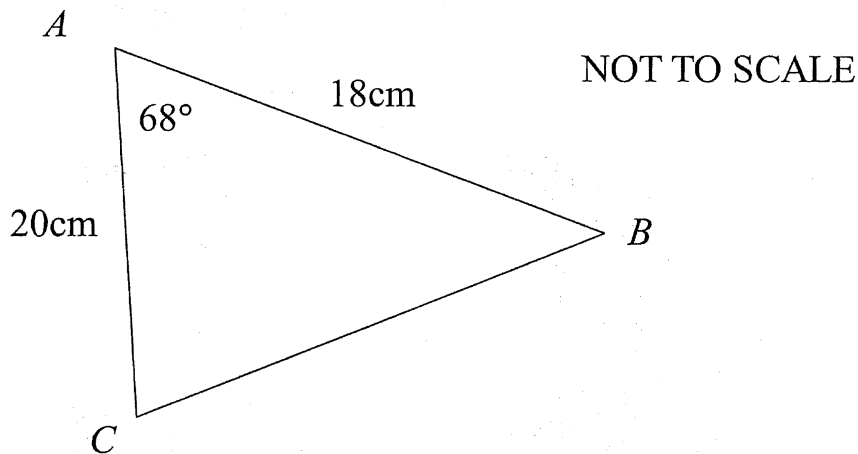
END OF QUESTION 11

Question 12 (12 marks) Answer on the appropriate page of your answer booklet

(a) Solve $\cot x = -\frac{\sqrt{3}}{3}$ for $0^\circ \leq x \leq 360^\circ$ 2

(b) Sketch the region defined by $y \geq x^2 + 2x - 8$ and $y < -2x - 8$ 3

(c) In the diagram, ABC is a triangle in which $\angle BAC = 68^\circ$, $AC = 20\text{cm}$ and $AB = 18\text{cm}$.



Copy or trace the diagram onto your worksheet.

i) Calculate the area of the triangle ABC , correct to 1 decimal place. 1

ii) Hence, or otherwise, find the length of the altitude of the triangle, from C to AB giving your answer correct to 1 decimal place. 1

(d) Town A is 4 km from town P and its bearing from town P is $030^\circ T$. Town B is due south of town A and 6km from town P .

(i) Draw a neat sketch that clearly illustrates the above information. 2

(ii) Find the distance between towns A and B , giving your answer correct to the nearest kilometre. 3

END OF QUESTION 12

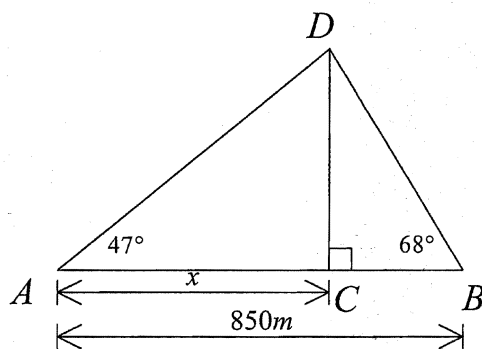
Question 13 (12 marks) Answer on the appropriate page of your answer booklet

- (a) If a and b are rational numbers and $a + 2b + \sqrt{3a - b} = 2 + 2\sqrt{5}$ find the value of a and the value of b . 3

- (b) (i) Show that $\frac{\cos A}{1 - \sin A} - \frac{\cos A}{1 + \sin A} = 2 \tan A$ 2

- (ii) Hence, or otherwise, solve $\frac{\cos A}{1 - \sin A} = 2 + \frac{\cos A}{1 + \sin A}$ for $0^\circ \leq A \leq 360^\circ$ 2

- (c) A and B are points directly opposite a hill DC. The angles of elevation of the top of the hill at A and B are 47° and 68° respectively and $AC = x$ m.



- (i) Using triangle ADC find an expression for DC in terms of x 1
- (ii) If the distance from A to B is 850m find the height of the hill correct to three significant figures. 2
- (d) Find the value of n if 4^n is equal to one quarter of 2^{88} 2

END OF QUESTION 13

Question 14 (12 marks) Answer on the appropriate page of your answer booklet

(a) Consider the curves $y = \frac{7}{x}$ and the line $y = 8 - x$.

(i) Find their points of intersection 2

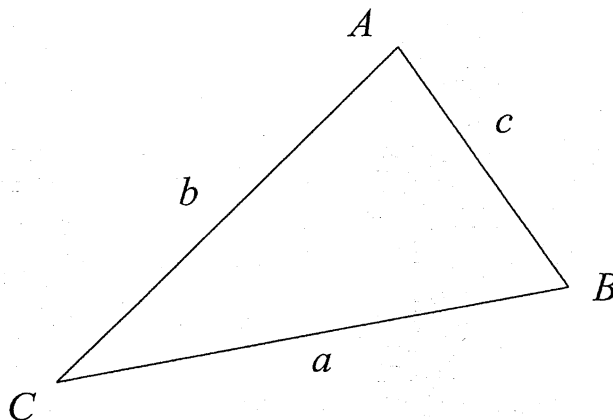
(ii) Sketch both curves on the same number plane. 2

(iii) Hence solve $\frac{7}{x} > 8 - x$ 2

(iv) What straight line should be drawn to intersect $y = \frac{2}{x}$ in order to solve $4x^2 - 3x - 2 = 0$? 1

(b) Solve $|6 - x| = 2x + 3$ 3

(c) Consider the following triangle below.



Show that if $\frac{1}{a+c} + \frac{1}{b+c} = \frac{3}{a+b+c}$ then $\angle ACB = 60^\circ$ 2

END OF EXAMINATION

1. $\frac{1-3\lambda}{2} \geq \frac{3}{2}$

$1-3\lambda \geq 3$

$-2 \geq 3\lambda$

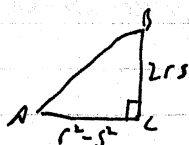
$\lambda \leq -\frac{2}{3}$

$\therefore (A)$

2. $5-\lambda \geq 0$ and $\lambda-4 \neq 0$
i.e. $\lambda \leq 5$ $\lambda \neq 4$

$\therefore (C)$

3.



$AB^2 = (r^2 - s^2)^2 + (2rs)^2$

$= r^4 + 2r^2s^2 + s^4$

$= (r^2 + s^2)^2$

$AB = r^2 + s^2$

$\cos A = \frac{r^2 - s^2}{r^2 + s^2}$

$\therefore (B)$

4. $\lambda-3=0$ or $2\lambda+1=0 \Rightarrow \lambda=-\frac{1}{2}$

$\therefore 2\lambda+1=0$ or $\lambda-3=0 \Rightarrow \lambda=3$

$\therefore 2\lambda+1=2 \times 3+1$

$=7$

$\therefore$ Or 7

$\therefore (B)$

5. shifted right $y=x^2$ (2 phases)

$\therefore (C)$

6. $\frac{k(k+1)(k+2)}{3} + \frac{3(k+1)(k+2)}{3}$

$= \frac{(k+1)(k+2)(k+3)}{3}$

(B)

7. $\frac{J}{K} = G - \frac{1}{T}$

$\frac{J}{K} - \frac{GK}{K} = -\frac{1}{T}$

$\frac{1}{T} = \frac{GK-J}{K}$

$T = \frac{K}{GK-J}$

$\therefore (A)$

8. $a^2 = a+L$

$a^3 = a^2 + La$

but $a^2 = a+L$

$\therefore a^3 = a+L+La$

$= 3a+L$

$\therefore (C)$

9. $\frac{\sin \alpha}{82} = \frac{\sin 26^\circ}{100}$

$\sin \alpha = \frac{82 \sin 26^\circ}{100}$

$\alpha = 21^\circ$ (nearest degree)

$\therefore \angle L = 47^\circ = (21+26)^\circ$

$\therefore \theta = 180 - 90 - 47$
 $= 43^\circ$

10. $\sqrt{7}\sqrt{\lambda} - \sqrt{3}\sqrt{\lambda} = 4$

$\sqrt{\lambda}(\sqrt{7}-\sqrt{3}) = 4$

$\sqrt{\lambda} = \frac{4}{\sqrt{7}-\sqrt{3}}$

$= \frac{4(\sqrt{7}+\sqrt{3})}{7-3}$

$\lambda = (\sqrt{7}+\sqrt{3})^2$

$= 10 + 2\sqrt{21}$

$\therefore (B)$

Section II

11 a) $4x - (2+x-x^2)$

$= 3x - 2 + x^2$

b) $\frac{2(\sqrt{5}+1)}{(\sqrt{5}-1)(\sqrt{5}+1)}$

$= \frac{2(\sqrt{5}+1)}{5-1}$

$= \frac{\sqrt{5}+1}{2}$

c) $4 - \frac{x}{3} = \frac{x}{2}$

$24 - 2x = 3x$

$5x = 24$

$x = \frac{24}{5}$ or $4\frac{4}{5}$ or 4.8

d) $3x^2 - 13x + 12 = 3x^2 - 9x - 4x + 12$

$= 3x(x-3) - 4(x-3)$

$= (3x-4)(x-3)$

② correct answer

① expands $(2-x)(x)$
simplifies by
expanding and negating

② correct answer

① multiplies by conjugate

② correct answer

① attempts to find
common denominator or
multiplies by 6

② correct answer

① progress towards
i.e. splits middle term/
finds one factor

e) $\sin \theta (2 \sin \theta - 1) = 0$
 $\sin \theta = 0$ or $2 \sin \theta = 1$
 $\theta = 0^\circ, 180^\circ, 360^\circ$ $\sin \theta = \frac{1}{2}$
 $\theta = 30^\circ, 150^\circ$
 $\therefore \theta = 0^\circ, 30^\circ, 150^\circ, 180^\circ, 360^\circ$

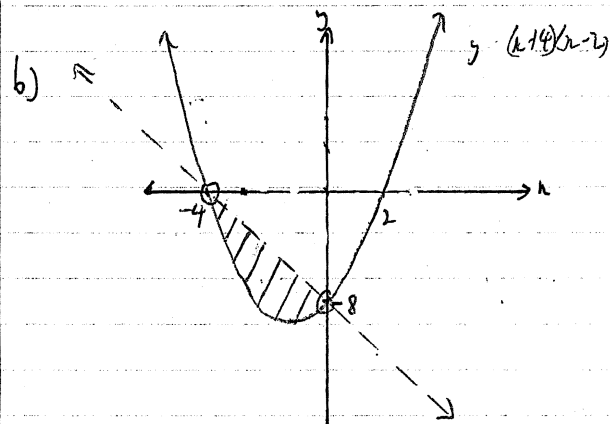
② correct solution
 ① solves $\sin \theta = 0$, $\sin \theta = \frac{1}{2}$ correctly.

f) If $a > 0$, $f(a) = a^2$, If $a < 0$, $f(a) = -(-a)^2$ ② correct solution
 $f(-a) = -(-a)^2 = -a^2$ with working for both cases
 $= -a^2$
 $= -f(a)$
 $f(-a) = (-a)^2 = a^2$ ① attempts to find $f(a)$ and compares to $f(a)$
 $= a^2$
 $= -f(a)$

If $a = 0$, $f(0) = 0$ $\therefore$ odd function

12 a) $\tan \lambda = \frac{-3}{4}$
 $\tan \lambda = -\frac{3}{4}$
 $\lambda = 120^\circ, 300^\circ$

② correct solution
 ① finds acute θ for $\tan$
 • finds one correct $\angle$
 • finds $150^\circ, 330^\circ$



③ correct solution
 ② both graphs correct
 • 1 region correct
 ① 1 graph correct (ignore open circles)

c) i) $A = \frac{1}{2} \times 20 \times 18 \sin 60^\circ$
 $= 166.9 \text{ cm}^2$

① correct answer

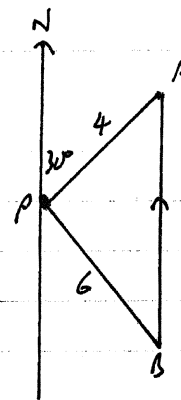
ii) $166.893... = \frac{18h}{2}$

① correct answer

$h = 18.5436...$

$\therefore \text{Altitude} = 18.5 \text{ cm 1dp}$

(d) (i)



① all information correct
 ① correct without vertices or all measurements

ii) $\angle PAB = 30^\circ$

by cosine rule

$6^2 = AB^2 + 4^2 - 2 \times 4 \times AB \cos 30^\circ$

③ correct solution

② solves cosine rule with 30° angle to form quadratic correctly solves incorrect gradient

$AB^2 - 4\sqrt{3}AB - 20 = 0$

$AB = \frac{4\sqrt{3} \pm \sqrt{48 + 80}}{2}$

① uses cosine rule

$= \frac{4\sqrt{3} \pm \sqrt{128}}{2}$

$= 9.1209...$

$= 9 \text{ km (nearest kilometre)}$

13 a) at $2b + \sqrt{3a-b} = 2 + \sqrt{20}$

at $2b = 2$ ①

$3a-b = 20$ ②

① $\times 3$ $3a+6b = 6$ ①A

①A - ② $7b = -14$

$b = -2$

sub in ① $a-4 = 2$

$a = 6$

$\therefore a = 6, b = -2$

③ correct answer

② finds a or b from correct simultaneous equations

• correctly finds a and b from incorrect equations

① correctly establishes simultaneous equations

① solves for a or b only from incorrect simultaneous equations

13 (b) (i) LHS $\frac{\cos A}{1-\sin A} \frac{1+\sin A}{1+\sin A} = \frac{\cos A(1+\sin A)}{(1-\sin A)(1+\sin A)}$
 $= \frac{\cos A + \cos A \sin A}{1-\sin^2 A} = \frac{\cos A + \cos A \sin A}{\cos^2 A}$
 $= \frac{2\cos A \sin A}{\cos^2 A} = \frac{2\sin A}{\cos A} = 2\tan A$
 $= \text{RHS}$

(A $\neq 90^\circ$) (2) correct solution
 (1) expresses with common denominator

(ii) $\frac{\cos A}{1-\sin A} = 2 + \frac{\cos A}{1+\sin A}$

(2) correct solution
 (1) uses (i) to find $2\tan A = 2$

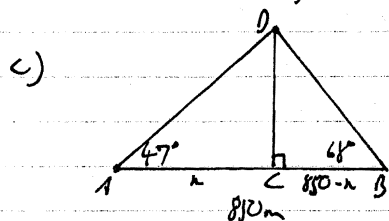
$\frac{\cos A}{1-\sin A} - \frac{\cos A}{1+\sin A} = 2$

$2\tan A = 2$

$\tan A = 1$

$A = 45^\circ, 180+45^\circ$

$\therefore A = 45^\circ, 225^\circ$



(2) correct solution
 (1) forms equation by equating OC in 2 triangles.
 or (1) finds AB or BC
 (1) finds OC from incorrect AB or BC

(i) In $\triangle AOC$
 (1) correct solution $\tan 47^\circ = \frac{OC}{x}$
 $OC = x \tan 47^\circ$

(ii) In $\triangle COB$
 $\tan 68^\circ = \frac{OC}{850-x}$
 $OC = (850-x) \tan 68^\circ$

$\therefore x \tan 47^\circ = (850-x) \tan 68^\circ$
 $x(\tan 47^\circ + \tan 68^\circ) = 850 \tan 68^\circ$
 $x = \frac{850 \tan 68^\circ}{\tan 47^\circ + \tan 68^\circ}$
 $x = 593.051...$
 $x = 593$

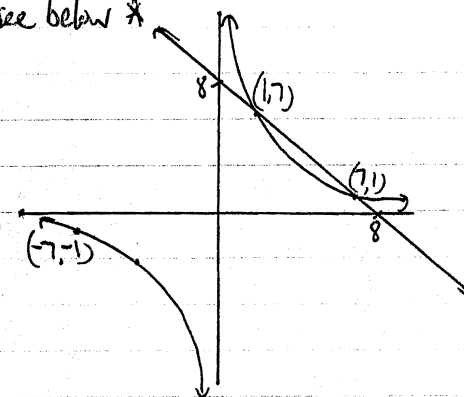
$47^\circ = \frac{OC}{x}$
 $OC = x \tan 47^\circ$
 $= 635.969x...$
 $= 636m$ (3 sig fig)

13 d) $4^n = \frac{2^{88}}{4}$
 $2^{2n} = \frac{2^{88}}{2^2}$
 $2^{2n} = 2^{86}$
 $2n = 86$
 $n = 43$

(2) correct solution
 (1) uses indices rules to simplify expression

14 (a) (i) see below *

14 (a) (ii)



(2) correct solution
 (1) one correct graph

* i) $\frac{7}{x} = 8-x$
 $7 = 8x - x^2$
 $x^2 - 8x + 7 = 0$
 $(x-7)(x-1) = 0$
 $x = 1, 7$

(2) correct solution
 (1) forms quadratic
 : finds 1 point of intersection

$\therefore$ pts of intersection (1, 7) and (7, 1)

(iii) $0 < x < 1$ or $x > 7$

(1) correct solution

(iv) $4x^2 - 3x - 2 = 0$
 $4x - 3 - \frac{2}{x} = 0$ ($\div x$)
 $4x - 3 = \frac{2}{x}$

(2) correct solution
 (1) divides by x

$\therefore y = 4x - 3$ is the required equation

$$b) \quad |6-x| = 2x+3$$

$$6-x = 2x+3 \quad \text{or} \quad 6-x = -2x-3$$

$$3 = 3x$$

$$x = 1$$

$$x = -9$$

no soln

$$\text{or } 2x+3 \geq 0$$

$$2x \geq -3$$

$$x \geq -\frac{3}{2}$$

③ correct solution
(with justification)

② $x = 1$ with no
justification/explanation
for other solution

④ $x = 1$ and $x = -9$

① $x = 1$ no justification

$\therefore x = 1$ is the only solution

$$c) \quad \frac{1}{a+c} + \frac{1}{b+c} = \frac{3}{a+b+c}$$

$$(a+b+c)(b+c) + (a+c)(a+b+c) = 3(a+c)(b+c)$$

$$ab+ac+cb^2+cb+bc+c^2+a^2+ab+ac+ac+bc+c^2 = 3(ab+ac+bc+c^2)$$

$$2ab+3ac+cb^2+3bc+2c^2+a^2$$

$$a^2+b^2-ab$$

$$= 3ab+3ac+3bc+3c^2$$

$$= c^2$$

Since $\cos 60^\circ = \frac{1}{2}$

$c^2 = a^2 + b^2 - 2ab \cos C$ which gives the cosine rule for
a triangle ABC with angle $C = 60^\circ$.

② correct solution

① simplifies to $c^2 = a^2 + b^2 - ab$
or equivalent.